



Low Strain Sonic Integrity Test/ Pile Integrity Test

Pile quality is often determined by performing a low strain integrity test directly after pile driving or within days after installation of cast-in-situ piles. Sonic Integrity Testing is a widely used method for testing the quality of concrete piles in a non-destructive way before they are incorporated in the final foundation plan.

The Profound SIT-series meets the constructions industry's demand for professional Sonic Integrity Testing of concrete piles. With the Profound SIT-series you can verify the pile length and check the integrity of foundation piles on defects after installation. Testing can point out which piles require further examination.

The SIT-series has been optimised for use in the field as well as for advanced interpretation and efficient management of the measurement data in the office. TheSIT models are robust, easily portable and user-friendly to operate at the construction site. One person can test numerous piles per hour.



Cross Hole Sonic Logging Test

The Cross Hole Ultrasonic/Sonic test uses the Cross-hole Sonic Logging (CSL) method to perform high-resolution quality control of deep foundations. The system uses an ultrasonic wave sent from a transmitter to a receiver pulled through water filled access tubes embedded in the concrete. The measured arrival time and energy are strongly dependent on the concrete quality.

Additional methods supported by the test are Single Hole Ultrasonic Testing and Tomography (two- and three-dimensional)

High Strain Pile Dynamic Analyzer/ Dynamic Load Test (PDA/ DLT)

Dynamic Load Testing (DLT) is a frequently used worldwide technique to evaluate the bearing capacity of piles. The Profound PDA/DLT-system is used for both DLT and for Pile Driving Analysis (PDA) as a powerful tool to control the pile driving process and reduce risks involved.

Pile Driving Analysis

During pile driving the performance of the hammer, the condition of the cushion, the behaviour of the pile and the driving resistance of the soil are accurately monitored and analysed.



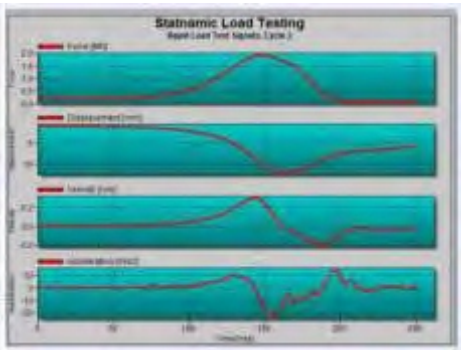
Dynamic Load Testing

The Dynamic Load Test (DLT) is a considerably faster and cost-effective alternative for a Static Load Test (SLT). With the Profound PDA/DLT-system the bearing capacity of several piles can be assessed in a single day. DLT is used for cast-in-place, prefabricated concrete piles and all types of steel piles



STATNOMIC™ Load Testing

Cost-effective method for determining the load bearing capacity of foundation piles. Statnamic is an innovative method for testing the load bearing capacity of foundation piles. Compared to static load testing: 90-95% less weight has to be loaded. Dependent upon the weight to be tested, Statnamic enables you to test up to 6 piles per day at much lower costs. Statnamic Load testing is performed in 4 stages. To prepare the test, the cylinder with pressure chamber is connected to the pile head and the reaction mass (only 5-10% of the load tested) is placed over the piston. The test starts with igniting the solid fuel propellant inside the pressure chamber, generating high-pressure gases and accelerating the reaction mass. At this moment, the actual loading of the pile takes place. The space caused by the upward movement of the reaction mass will be filled by gravel. The gravel catches the reaction mass and transfers the impact forces to the subsoil. The applied force, displacement and acceleration are directly monitored. All signals are measured by the data acquisition system whereby the mobilized capacity and static load displacement behaviour can immediately be presented in a load-displacement diagram.



TNOWAVE – Wave Equation Program

TNOWAVE represents a group of wave equation application programs and is a powerful simulation tool to predict pile behaviour. The TNOWAVE applications are suitable for simulating pile-soil interaction under dynamic conditions.

TNOWAVE is based on the one dimensional stress wave theory and allows the modelling of:

- All types of vibratory and impact hammers with anvils and cushions
- Piles with cross sectional variations and cracks
- Tubular steel piles, open ended or closed
- Shaft friction and toe resistance

TNOWAVE has the following application modules:

- PDPWAVE for impact hammer and vibratory hammer driveability studies
- SITWAVE to determine local pile defects from sonic integrity testing signals by signal matching
- DLTWAVE to determine pile bearing capacity from a dynamic load test by signal matching